SplitSmart ■ — Executive & Technical Report

Web Brawl GC 2026 Submission · 100% Client-Side AI Receipt Scanner & Fair Debt Simplification

GitHub Repository: https://github.com/Sandesh187/splitsmart	**Live Deployment:** https://sandesh187.github.io/splitsmart/
Stack: React 18, TypeScript 5, Vite 6, Tailwind CSS, Tesseract.js, jsPDF	**Privacy Guarantee:** 100% Client-Side Web Worker inference. Zero server uploads.

COMPREHENSIVE FEATURE COMPLIANCE (TIERS 1, 2 & 3)

Tier 1: Core Requirements (100% Complete)
- **Groups & Members:** Full CRUD, color avatars, UPI IDs.
- **Receipt Capture:** Device camera & image upload (JPG/PNG).
- **Structured OCR:** Items, Qty, Unit, Line Total, CGST/SGST.
- **Review & Correction:** Side-by-side view, low-confidence badges.
- **Total Validation:** Real-time sum alert & 1-click reconcile.
- **Manual Expenses:** Log expenses without receipt across 4 modes.
- **Balances View:** Real-time net debt calculation (+ owed / - owes).

Tier 2: Advanced Features (100% Complete)
- **Itemized Assignment:** Tap-to-assign avatars; unassigned guard.
- **4 Split Modes:** Equal, Exact, Percentage (100%), Shares (2:1:1).
- **Proportional Tax:** Tax weighted by consumed subtotal (not equal).
- **Paise-Exact Rounding:** Integer paise; **zero paisa drift** rule.
- **Debt Simplification:** Greedy min-cash-flow algorithm (<= N-1).
- **Auto-Categorization:** Lazy zero-shot ML + fast heuristic.
- **Persistence:** Full state survives reload via LocalStorage.

Tier 3: Brownie Points & Innovation (100% Complete)
- **Instant UPI Settle-Up:** NPCI upi://pay deep links & scannable QR codes for GPay/PhonePe/Paytm.
- **Spending Analytics:** Visual charts: Category breakdown & consumption vs paid metrics.
- **Serverless Group Sharing:** JSON import/export & compressed base64 shareable URL state.

Additional Tier 3 Highlights & Testing
- **PWA & Offline:** Web App Manifest, mobile responsive (360px-1440px).
- **Recurring Bills:** Automated monthly/weekly items (Wi-Fi, rent).
- **Reports Export:** 1-click download of PDF reports & CSV summaries.
- **In-App Test Suite:** 1-click runner for all 9 Section 10 test cases.

CORE ALGORITHMS & MATHEMATICAL FORMULATIONS

1. Proportional Tax & Discount Formula:
Tax is allocated proportionally to member item subtotal:
ShareTax(u) = floor([Subtotal(u) / SubtotalTotal] × TaxTotal) + RemainderPaisa
Verified: On ■1,000 bill with ■100 tax, ■600 items pays exactly ■60 tax.

2. Paise-Exact Rounding Rule (0 Drift):
All math uses integer paise (■1 = 100 paise). Remainder diff = TotalPaise - Sum(Floored). Top diff members sorted by fractional remainder absorb exactly 1 paisa.
Guarantees: Sum of individual shares == Bill Total to the paisa.

3. Greedy Minimum Cash Flow Debt Simplification:
Computes net balance: *Net(u) = TotalPaid(u) - TotalOwed(u) + SettledDelta(u)* with invariant Sum(Net) == 0.
Iteratively matches largest debtor with largest creditor for amount *T = min(|maxDebtor|, maxCreditor)*.
Reduces N×(N-1) potential pairwise debts down to at most (N-1) clean transactions.

4. Section 10 Robustness Test Verification:
- **Case 6 (■100 split 3 ways):** Exactly ■33.34, ■33.33, ■33.33 (reconciles).
- **Case 9 (5 members, 10 expenses):** Simplified to 4 payments.
- **Cases 1-5, 7-8:** Rotated 90°, faded contrast, missing total handled.

Unit Tests: 8/8 automated Vitest tests pass in 940ms.

OFFICIAL SUBMISSION DESCRIPTION (GOOGLE FORM READY)

"SplitSmart is a client-side bill-splitting web app built for Web Brawl GC 2026. Features include Tesseract.js in-browser OCR with contrast enhancement and 90° rotation preprocessing; lazy-loaded auto-categorization (Xenova/mobilebert-uncased-mnli with heuristic fallback); proportional tax and discount allocation; paise-exact rounding reconciliation (zero drift); greedy minimum cash flow debt simplification; instant UPI settle-up with NPCI deep links and QR codes; visual spending analytics; recurring expense automation; PDF/CSV export; serverless group sharing via URL; PWA offline support; and an interactive in-app test runner for all 9 Section 10 robustness cases."